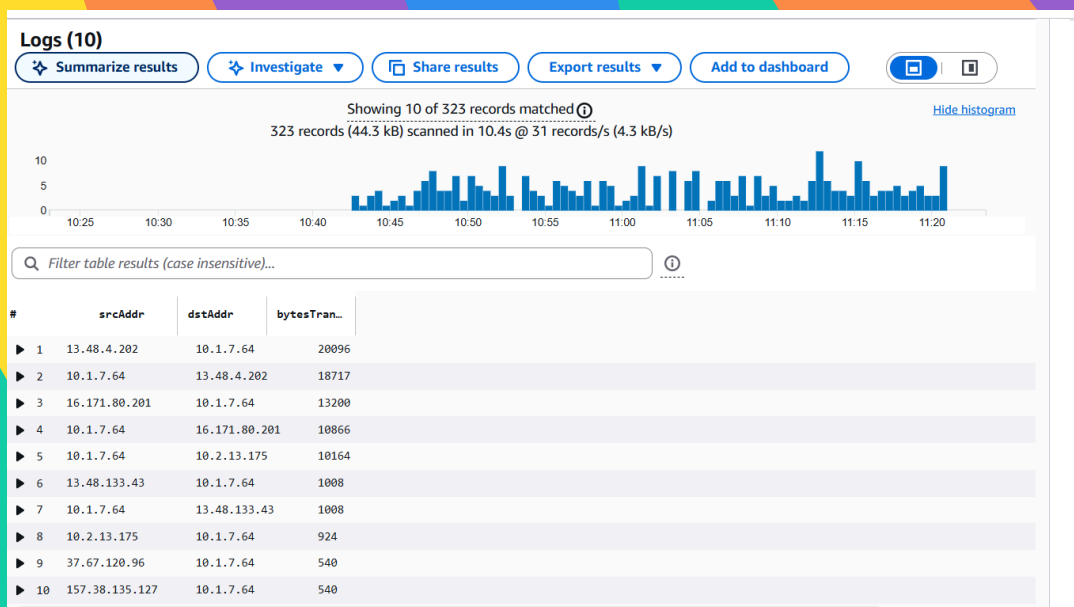




VPC Monitoring with Flow Logs



MHM Aslam



A circular profile picture of a man with short dark hair, wearing a blue and white sports jersey, standing in front of a green background with some text and graphics.

Introducing Today's Project!

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) is a service that lets you create a private, isolated section of the AWS cloud where you can launch resources like EC2 instances. It is useful because it gives you full control over your network environment, including IP address ranges, subnets, route tables, security groups, and network gateways, allowing you to build a secure and scalable network tailored to your application needs.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create two separate VPCs (NextWork-1 and NextWork-2) each with a public subnet, launch EC2 instances inside them, set up a VPC peering connection between the VPCs, update route tables to direct traffic through the peering connection, and enable VPC Flow Logs to monitor and analyze the network traffic between the instances.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was that even though both instances had public IPs and could ping each other over the internet, they could not communicate using their private IP addresses until I added the correct routes to the VPC peering connection. I assumed the peering connection alone would be enough, but the route tables were the missing piece.



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This project took me...

This project took me about 45 minutes to an hour, including reading the instructions, creating the VPCs, launching instances, setting up the peering connection, enabling flow logs, and running the queries in Logs Insights.



In the first part of my project...

Step 1 - Set up VPCs

In this step, I will create two VPCs from scratch using the VPC wizard because I need a setup where I can later enable flow logs and test monitoring between two separate networks, just like we did in the VPC peering project.

Step 2 - Launch EC2 instances

In this step, I will launch one EC2 instance in each VPC, because later we need these instances to send data to each other through the VPC peering connection, which will create network traffic for us to monitor using flow logs.

Step 3 - Set up Logs

In this step, I will set up VPC Flow Logs for my VPCs and also create a destination like Amazon CloudWatch Logs or S3 to store the log records, because this allows me to capture and track all inbound and outbound network traffic for monitoring and troubleshooting later.

Step 4 - Set IAM permissions for Logs

In this step, I will create an IAM role and policy that gives VPC Flow Logs permission to write logs and send them to CloudWatch Logs, because without these permissions, the flow logs cannot store the network traffic records I need for monitoring.



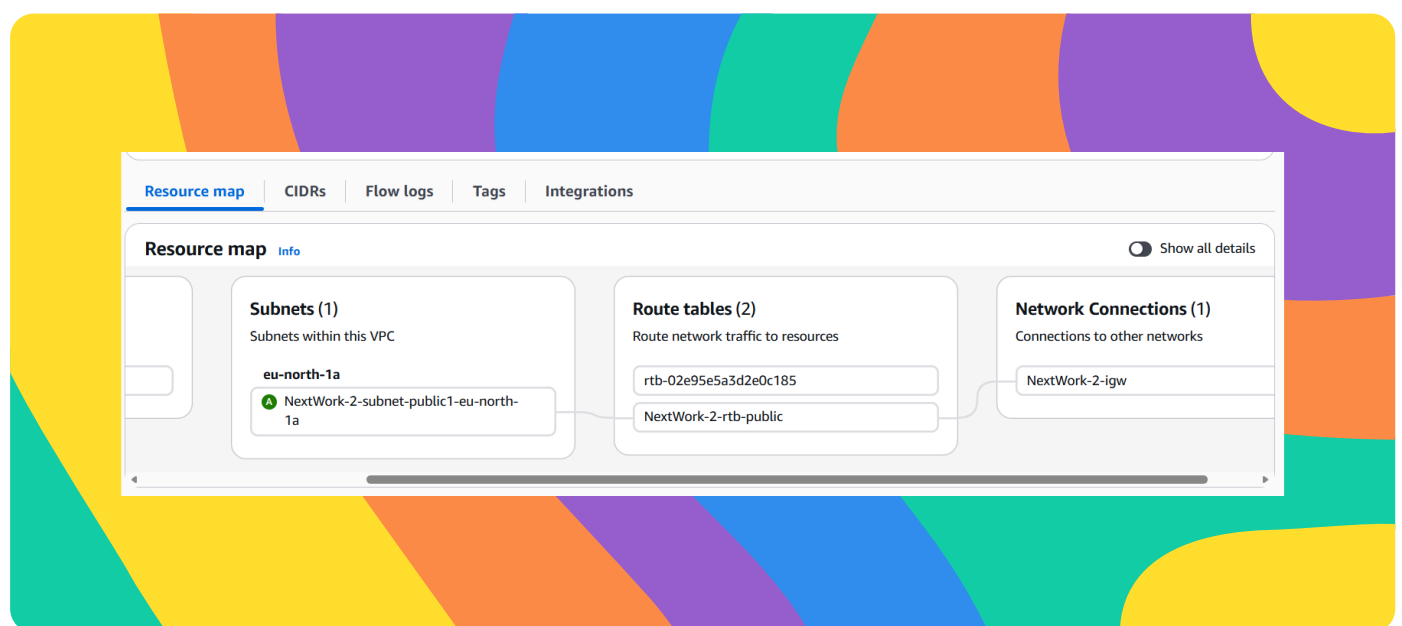
Multi-VPC Architecture

I started my project by launching two VPCs. For each VPC, I created one public subnet and no private subnets. So in total, I created two public subnets.

The CIDR blocks for VPCs 1 and 2 are 10.1.0.0/16 and 10.2.0.0/16. They have to be unique because if the IP address ranges overlap, it will cause routing conflicts and connectivity issues between the two VPCs.

I also launched EC2 instances in each subnet

My EC2 instances' security groups allow All ICMP - IPv4 traffic from source 0.0.0.0/0. This is because we will be doing ping tests later, and those tests require the security group to accept ICMP traffic coming from any IP address, not just from the other VPC.





Logs

Logs are like a diary for your computer systems. They record everything that happens, from users logging in to errors popping up. It's the go-to place to understand what's going on with your systems, troubleshoot problems, and keep an eye on who's doing what.

Log groups are like big folders in AWS where you keep related logs together. Logs from the same source or application usually go into the same log group, and log data is created and saved in the region where it was made.

Successfully created flow log for the following resource:
vpc-0b1b4cc011f6a14ea

fl-0c1cfc2d5e40f878d / NextWorkVPCFlowLog

Actions

Details

Flow Log ID fl-0c1cfc2d5e40f878d	Destination Type cloud-watch-logs	Traffic Type All	File Format -
Name NextWorkVPCFlowLog	Destination Name NextWorkVPCFlowLogsGroup	Max Aggregation Interval 1 minute	Hive Compatible Partitions -
State Active	IAM Role arn:aws:iam::799227258538:role/NextWorkVPCFlowLogsRole	Log Format Default	Partition Logs -
Creation Time Friday, May 15, 2026 at 16:11:04 GMT+5:30	Cross Account IAM Role -	Tag field specifications -	

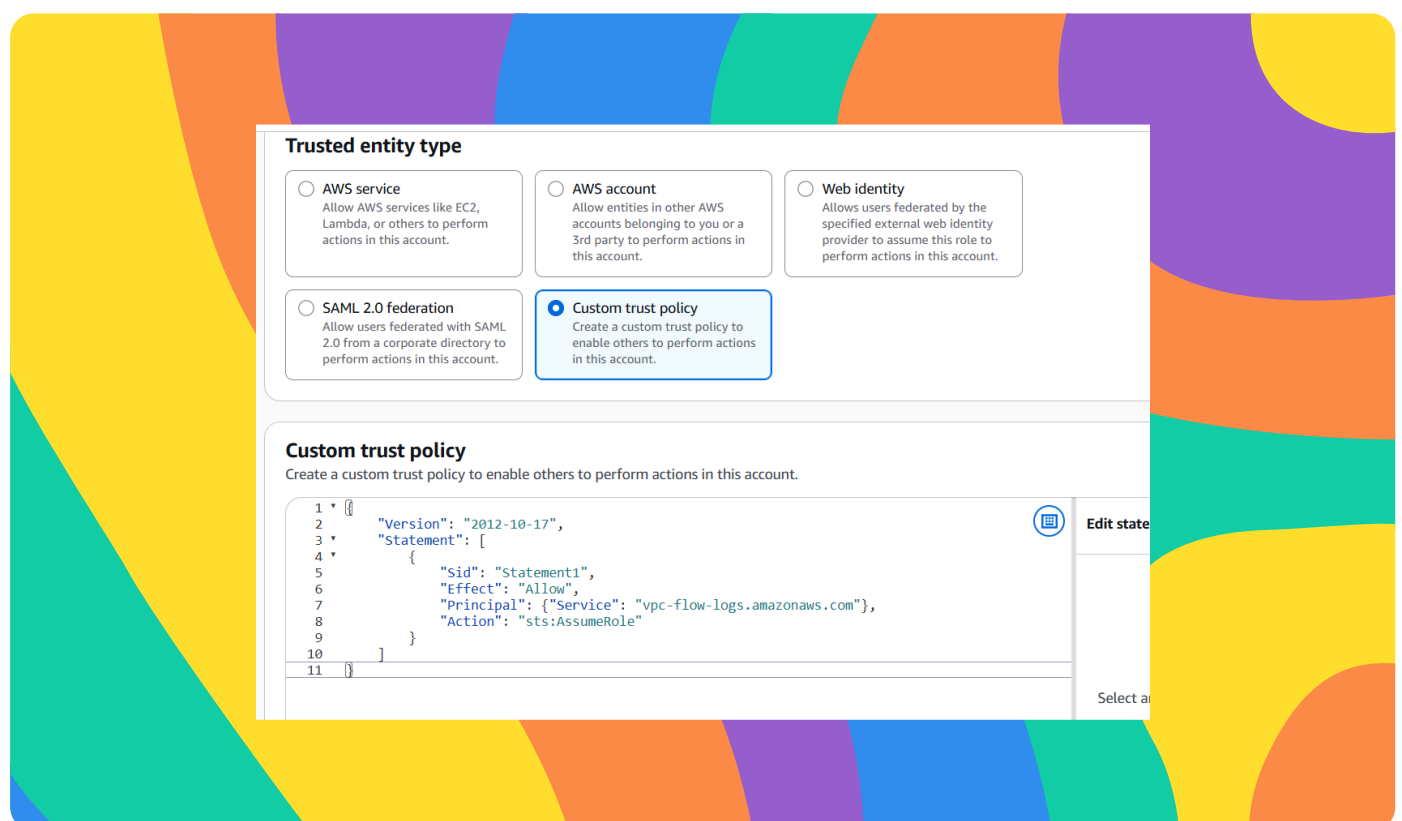


IAM Policy and Roles

I created an IAM policy because VPC Flow Logs do not have the default permission to record logs and store them in CloudWatch Log groups. This policy gives Flow Logs the access it needs to create log groups, create log streams, and send log data to them, so I can actually capture and monitor my network traffic.

I also created an IAM role because I needed a way to attach the IAM policy to VPC Flow Logs. The role with its custom trust policy makes sure that only VPC Flow Logs can use those permissions to write logs to CloudWatch.

A custom trust policy is a special type of policy that narrowly defines which specific service, user, or role is allowed to assume (use) an IAM role. It acts like a VIP list—only the entity you pinpoint in the policy can take on that role's permissions.



The screenshot shows the AWS IAM console interface for creating a custom trust policy. The background is a colorful abstract pattern. The main content area is divided into two sections: 'Trusted entity type' and 'Custom trust policy'.

Trusted entity type

- ☐ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☒ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

```
1  {
2    "Version": "2012-10-17",
3    "Statement": [
4      {
5        "Sid": "Statement1",
6        "Effect": "Allow",
7        "Principal": {"Service": "vpc-flow-logs.amazonaws.com"},
8        "Action": "sts:AssumeRole"
9      }
10   ]
11 }
```

On the right side of the console, there are buttons for 'Edit state' and 'Select a...'.

A circular profile picture of a person with dark hair, wearing a blue shirt, standing in front of a colorful background.

In the second part of my project...

Step 5 - Ping testing and troubleshooting

In this step, I will use Instance 1 (in VPC 1) to send test messages to Instance 2 (in VPC 2) using ping or similar commands, because this generates actual network traffic across the VPC peering connection, allowing me to check if my flow logs are recording the traffic properly and also confirm that the peering connection is working correctly.

Step 6 - Set up a peering connection

In this step, I will create a VPC peering connection between VPC 1 and VPC 2 and update the route tables in both VPCs, because this will allow the two EC2 instances to communicate directly using their private IP addresses instead of going through the public internet.

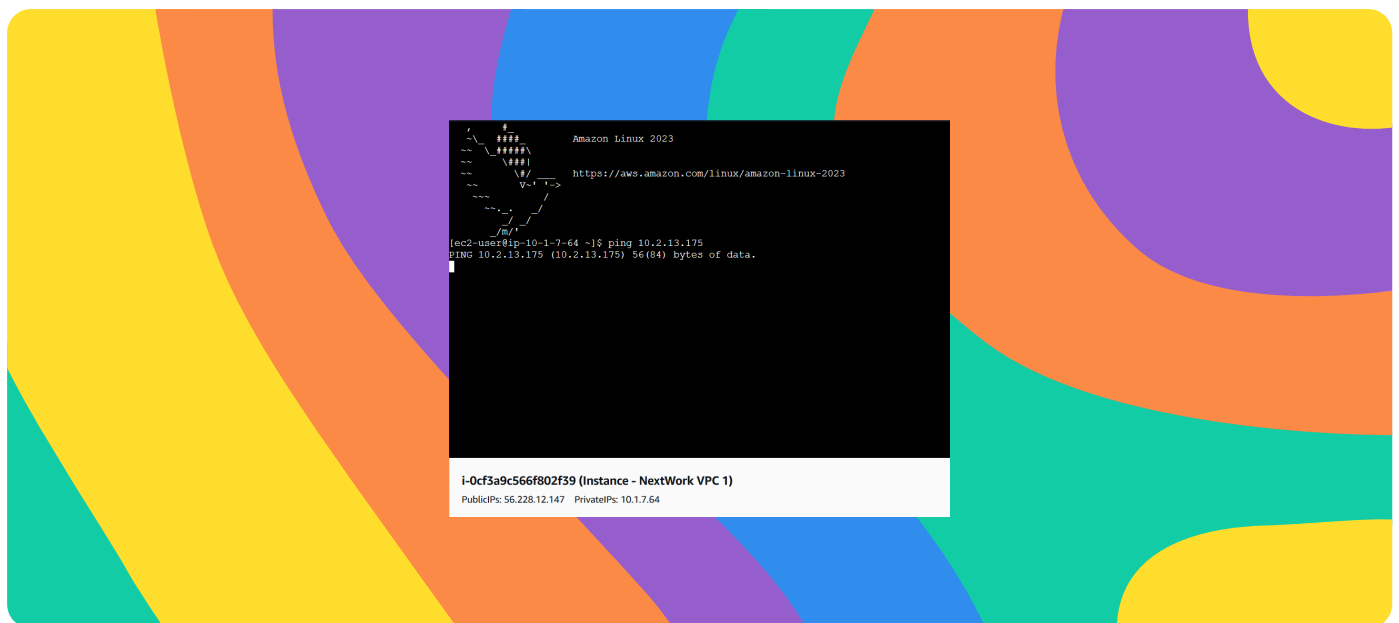
Step 7 - Analyze flow logs

In this step, I will review and analyze the flow logs recorded from VPC 1's public subnet because this allows me to see what network traffic was actually captured, verify that my flow log setup is working correctly, and gain insights into which connections were accepted or rejected.



Connectivity troubleshooting

My first ping test between my EC2 instances had no replies, which means the two instances could not communicate with each other. This could be due to missing or incorrect VPC peering connection, missing routes in the route tables, or security groups not allowing ICMP traffic between the VPCs.



I could receive ping replies if I ran the ping test using the other instance's public IP address, which means the traffic is travelling over the public internet instead of through the VPC peering connection. This confirms that the instances have internet access and public IPs enabled, but does not test whether the VPC peering connection is working properly.

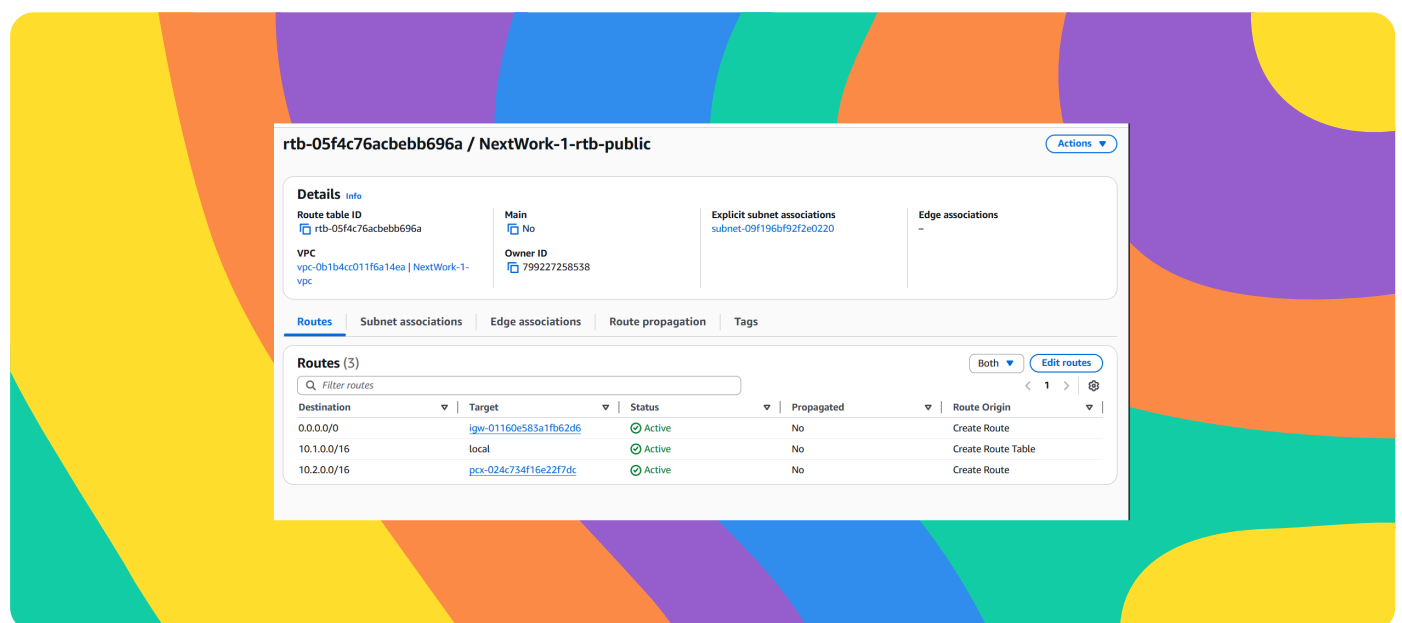


Connectivity troubleshooting

Looking at VPC 1's route table, I identified that the ping test with Instance 2's private address failed because there was no direct route to VPC 2's CIDR block (10.2.0.0/16) via the VPC peering connection. The only route was 0.0.0.0/0 through the internet gateway, which only works for public IP addresses and sends traffic over the public internet, not the private peering link.

To solve this, I set up a peering connection between my VPCs

I also updated both VPCs' route tables so that traffic destined for the other VPC's CIDR block is directed to the VPC peering connection instead of the internet gateway. Without these routes, the instances would not know how to reach each other using their private IP addresses.



rtb-05f4c76acbebb696a / NextWork-1-rtb-public

Details

Route table ID: rtb-05f4c76acbebb696a

VPC: vpc-0b1b4cc011f6a14ea | NextWork-1-vpc

Main: No

Owner ID: 799227258538

Explicit subnet associations: subnet-09f196bf92f2e0220

Edge associations: -

Routes (3)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-01160e583a1fb62d6	Active	No	Create Route
10.1.0.0/16	local	Active	No	Create Route Table
10.2.0.0/16	pcx-024c734f16e22f7dc	Active	No	Create Route



Connectivity troubleshooting

I received ping replies from Instance 2's private IP address! This means the VPC peering connection is working properly, and the two instances can now communicate directly using their private IP addresses without traffic going through the public internet.

```
[ec2-user@ip-10-1-7-64 ~]$ ping 10.2.13.175
PING 10.2.13.175 (10.2.13.175) 56(84) bytes of data.
64 bytes from 10.2.13.175: icmp_seq=1 ttl=127 time=0.449 ms
64 bytes from 10.2.13.175: icmp_seq=2 ttl=127 time=0.264 ms
64 bytes from 10.2.13.175: icmp_seq=3 ttl=127 time=0.264 ms
64 bytes from 10.2.13.175: icmp_seq=4 ttl=127 time=0.257 ms
64 bytes from 10.2.13.175: icmp_seq=5 ttl=127 time=0.217 ms
64 bytes from 10.2.13.175: icmp_seq=6 ttl=127 time=0.316 ms
64 bytes from 10.2.13.175: icmp_seq=7 ttl=127 time=0.215 ms
64 bytes from 10.2.13.175: icmp_seq=8 ttl=127 time=0.335 ms
64 bytes from 10.2.13.175: icmp_seq=9 ttl=127 time=0.276 ms
64 bytes from 10.2.13.175: icmp_seq=10 ttl=127 time=0.296 ms
64 bytes from 10.2.13.175: icmp_seq=11 ttl=127 time=0.270 ms
^C
--- 10.2.13.175 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10365ms
rtt min/avg/max/mdev = 0.215/0.287/0.449/0.061 ms
[ec2-user@ip-10-1-7-64 ~]$
```

i-0cf3a9c566f802f39 (Instance - NextWork VPC 1)

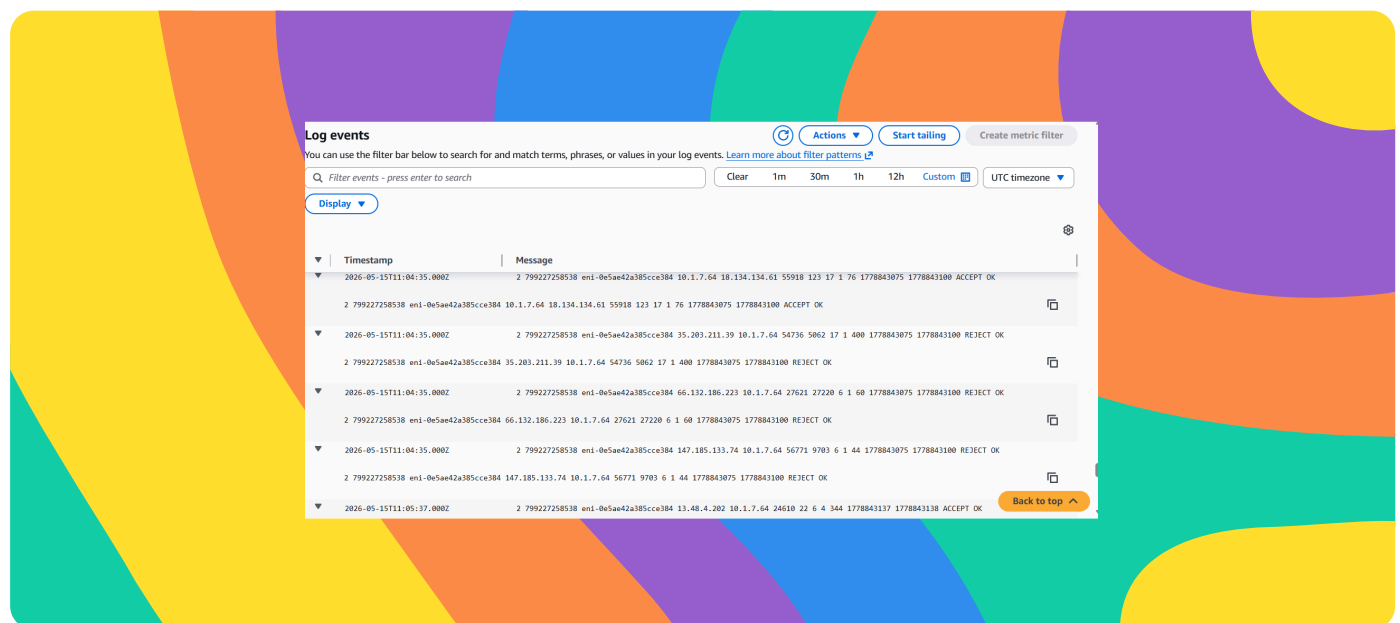
PublicIPs: 56.228.12.147 PrivateIPs: 10.1.7.64



Analyzing flow logs

Flow logs tell us about the source IP address, destination IP address, number of bytes transferred, number of packets sent, the protocol used (like TCP or ICMP), the port number, and whether the traffic was accepted (ACCEPT) or rejected (REJECT) by your security settings.

For example, the flow log I've captured tells us that 344 bytes of data were sent successfully from the IP address 18.237.140.165 to the instance with private IP 10.1.5.112 using TCP protocol on port 22, with 4 packets transferred, and the traffic was marked as "ACCEPT". This matches EC2 Instance Connect's IP range in the Oregon region, meaning someone connected to the instance securely via EC2 Instance Connect.



The screenshot shows the AWS CloudWatch 'Log events' console. The interface includes a search bar, a filter bar with options like 'Clear', '1m', '30m', '1h', '12h', 'Custom', and 'UTC timezone', and a table of log events. The table has two columns: 'Timestamp' and 'Message'. The messages contain detailed flow log data, including source and destination IP addresses, ports, protocols, and the result (ACCEPT or REJECT).

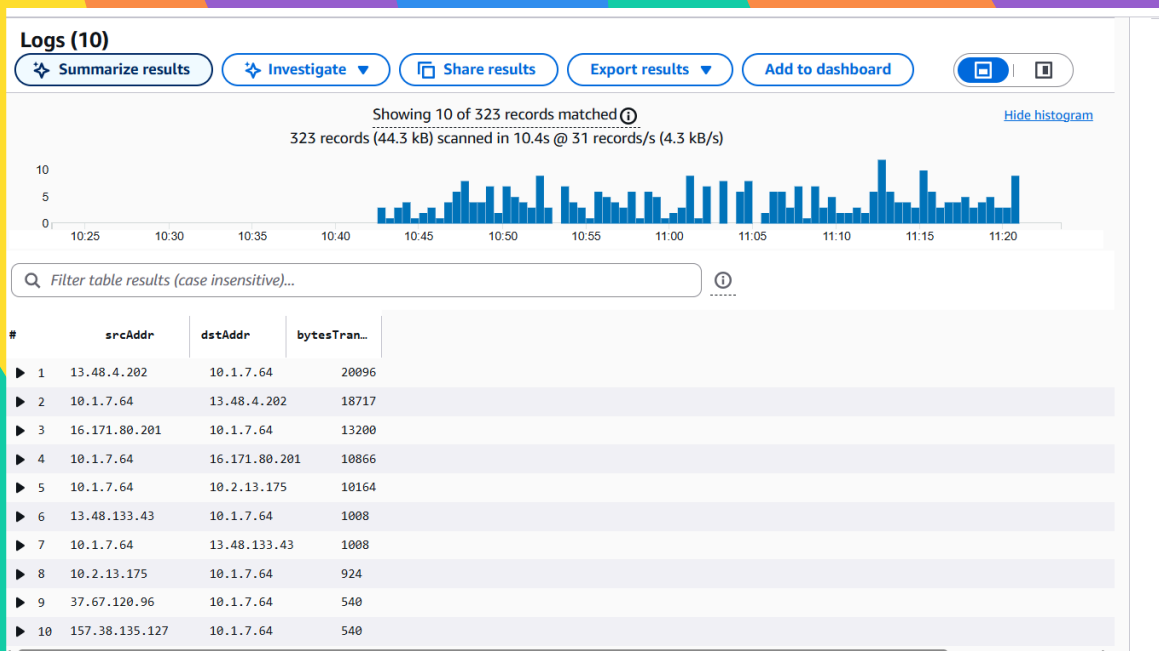
Timestamp	Message
2026-05-15T11:04:35.000Z	2 799227258538 eni-0e5ae42a385cce384 10.1.7.64 18.134.134.61 55918 123 17 1 76 1778843075 1778843100 ACCEPT OK
2 799227258538 eni-0e5ae42a385cce384 10.1.7.64 18.134.134.61 55918 123 17 1 76 1778843075 1778843100 ACCEPT OK	
2026-05-15T11:04:35.000Z	2 799227258538 eni-0e5ae42a385cce384 35.203.211.39 10.1.7.64 54736 5062 17 1 400 1778843075 1778843100 REJECT OK
2 799227258538 eni-0e5ae42a385cce384 35.203.211.39 10.1.7.64 54736 5062 17 1 400 1778843075 1778843100 REJECT OK	
2026-05-15T11:04:35.000Z	2 799227258538 eni-0e5ae42a385cce384 66.132.186.223 10.1.7.64 27621 27220 6 1 60 1778843075 1778843100 REJECT OK
2 799227258538 eni-0e5ae42a385cce384 66.132.186.223 10.1.7.64 27621 27220 6 1 60 1778843075 1778843100 REJECT OK	
2026-05-15T11:04:35.000Z	2 799227258538 eni-0e5ae42a385cce384 147.185.133.74 10.1.7.64 56771 9703 6 1 44 1778843075 1778843100 REJECT OK
2 799227258538 eni-0e5ae42a385cce384 147.185.133.74 10.1.7.64 56771 9703 6 1 44 1778843075 1778843100 REJECT OK	
2026-05-15T11:05:37.000Z	2 799227258538 eni-0e5ae42a385cce384 13.48.4.202 10.1.7.64 24610 22 6 4 344 1778843137 1778843138 ACCEPT OK



Logs Insights

Logs Insights is a CloudWatch feature that analyzes your logs. In Logs Insights, you use queries to filter, process and combine data to help you troubleshoot problems or better understand your network traffic.

I ran the query "Top 10 byte transfers by source and destination IP addresses". This query analyzes all the captured flow logs and returns the ten pairs of source and destination IP addresses that transferred the most bytes between them, helping to identify the busiest traffic routes or unusually large data transfers in the network.





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